

Clearance / Creepage Design Verification Report

Project Name	Test Project	Project No.	—
Prepared By	Test Engineer	Date	2026-09-15
Reference Standard	IEC 60664-1	Verification Method	Selected per dimension (default: RSS 3σ)

CELL

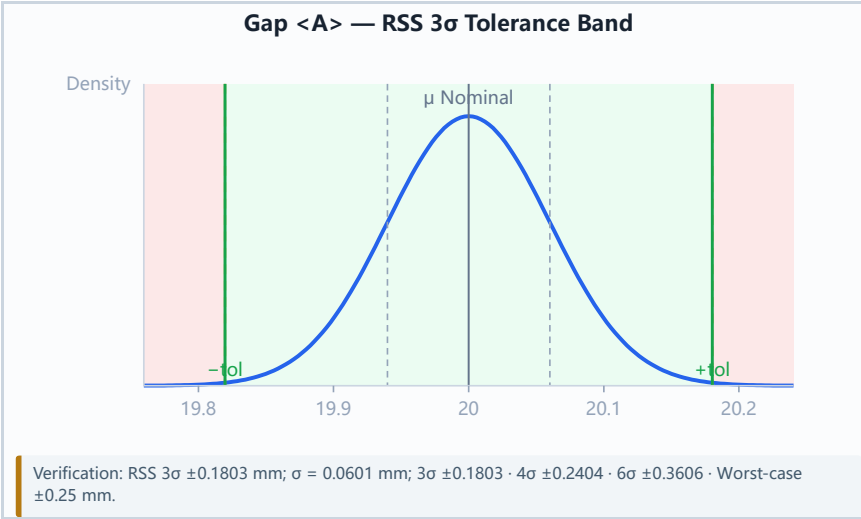
Cell Level Verification - PASS

Working Voltage	4.2 V	Altitude	2000 m	Pollution Degree	2	Material Group	IIIa
Uimp	1.426 kV	Required Clearance	0.5 mm	Required Creepage	0.4 mm	Altitude Factor	1

#	Critical Dimension	Image	Category	Nominal / mm	Tolerance / mm	Minimum / mm	Required / mm	Notes	Result
1	Gap <A>		Clearance	20	±0.1803 (RSS 3σ) 3σ ±0.1803 · 4σ ±0.2404 · 6σ ±0.3606 · Worst-case ±0.25	19.8197	0.5	—	PASS

Gap <A> — Dimension Chain and Tolerance Distribution

Dimension Name	Size / mm	Tolerance / mm	Type	Direction	Factor	Contrib. / mm
Main dimension	20	0.15	Size tolerance ±t	Positive	1	0.15
Profile	0	0.2	Profile t	Negative	0.5	0.1



MODULE

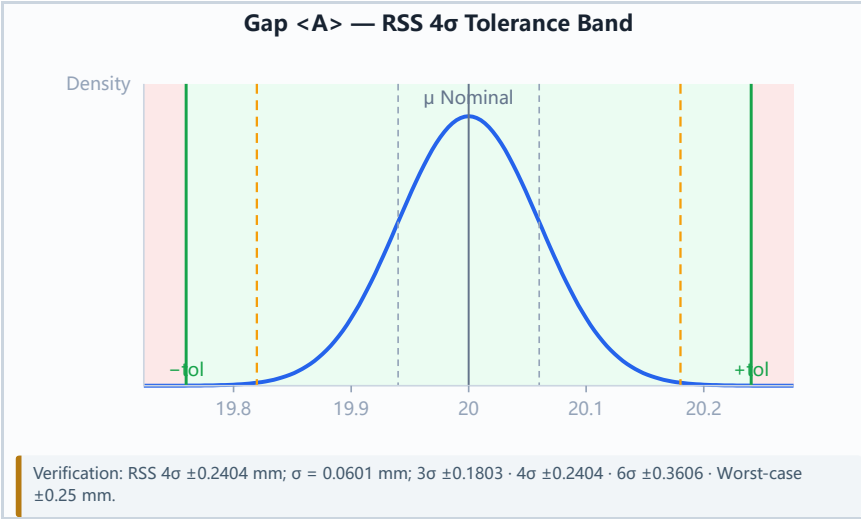
Module Level Verification - PASS

Working Voltage	60 V	Altitude	2000 m	Pollution Degree	3	Material Group	IIIa
Uimp	1.584 kV	Required Clearance	1 mm	Required Creepage	2 mm	Altitude Factor	1

#	Critical Dimension	Image	Category	Nominal / mm	Tolerance / mm	Minimum / mm	Required / mm	Notes	Result
1	Gap <A>		Creepage	20	±0.2404 (RSS 4σ) 3σ ±0.1803 · 4σ ±0.2404 · 6σ ±0.3606 · Worst-case ±0.25	19.7596	2	—	PASS

Gap <A> — Dimension Chain and Tolerance Distribution

Dimension Name	Size / mm	Tolerance / mm	Type	Direction	Factor	Contrib. / mm
Main dimension	20	0.15	Size tolerance ±t	Positive	1	0.15
Profile	0	0.2	Profile t	Negative	0.5	0.1



PACK

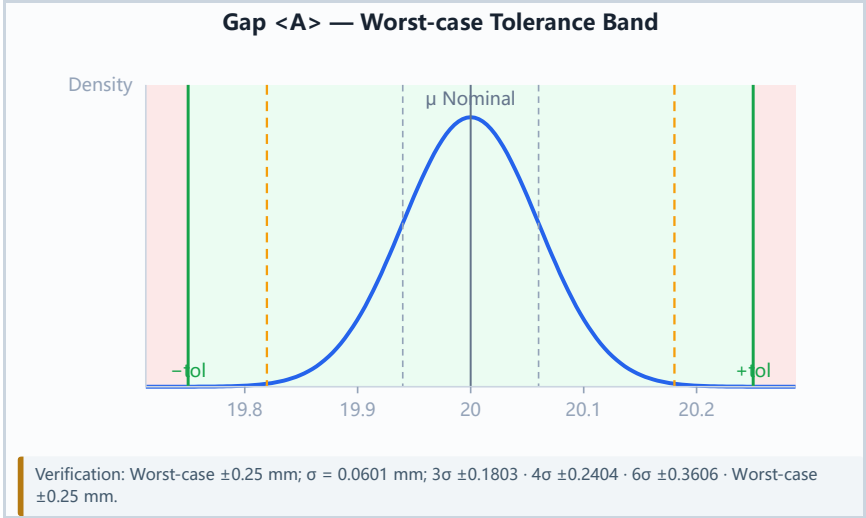
Pack Level Verification - PASS

Working Voltage	800 V	Altitude	2000 m	Pollution Degree	3	Material Group	IIIa
Uimp	3.677 kV	Required Clearance	3 mm	Required Creepage	12.5 mm	Altitude Factor	1

#	Critical Dimension	Image	Category	Nominal / mm	Tolerance / mm	Minimum / mm	Required / mm	Notes	Result
1	Gap <A>		Clearance	20	±0.25 (Worst-case) $3\sigma \pm 0.1803 \cdot 4\sigma$ $\pm 0.2404 \cdot 6\sigma$ $\pm 0.3606 \cdot \text{Worst-case} \pm 0.25$	19.75	3	—	PASS

Gap <A> — Dimension Chain and Tolerance Distribution

Dimension Name	Size / mm	Tolerance / mm	Type	Direction	Factor	Contrib. / mm
Main dimension	20	0.15	Size tolerance ±t	Positive	1	0.15
Profile	0	0.2	Profile t	Negative	0.5	0.1



Verification Summary

Overall result: **PASS**. Dimension chains default to RSS 3σ ; RSS 4σ , RSS 6σ or worst-case may be selected per dimension. RSS assumes independent, approximately normal contributors. Worst-case adds absolute tolerance contributions. Manually overridden results are identified in the detail table and take precedence over calculated results.

- Cell Level: PASS; completed 1/1 checks; failed 0 checks.
- Module Level: PASS; completed 1/1 checks; failed 0 checks.
- Pack Level: PASS; completed 1/1 checks; failed 0 checks.